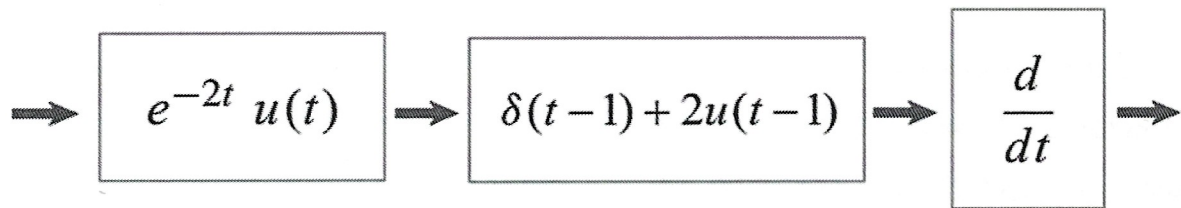


1. (5 points) What is the equivalent impulse response of the cascade of systems shown below?



2. (5 points) Calculate the integral $y(t) = \int_{-3}^{-1} t^5 \delta(t+2) dt + \int_0^2 t^5 \delta(t-4) dt$

3. **(5 points)** Explain how any signal can be represented as a sum of an even symmetric signal and an odd symmetric signal. Identify the odd and even components $x_o(t)$ and $x_e(t)$, respectively, of the signal $x(t) = 1 - 2\sin^2(2t)$.

4. **(5 points)** What is average power? If the average power of a signal $x(t)$ is P , what is the average power of the signal $2x(3t)$?

5. **(5 points)** Consider the signal $x(t) = \sin(\pi^2 t)$. What is the period of this signal? Now suppose that you time scale the signal by a factor of a : $y(t) = x\left(\frac{t}{a}\right)$. By what factor a would you scale the signal so that $y(n) = x\left(\frac{n}{a}\right) = 0$, for all integers n ?

6. **(5 points)** Is the system $y(t) = x(t - 2) + x(2 - t)$ linear time invariant? Explain your reasons.

7. **(5 points)** How do you define a bounded signal? Derive the condition an LTI system satisfies to be BIBO stable?

8. **(5 points)** Compute the convolution $y(t) = u(t - 1) * (\delta(t) + 2u(t) + 4r(t - 1))$ without computing any integrals.

9. **(10 points)** Calculate the output $y(t)$ of an LTI system whose impulse response is $h(t)$ when it is provided the input $x(t)$, where: $x(t) = te^{at}u(t)$ and $h(t) = e^{bt}u(t)$, $a \neq b$.